North South University

Department of Electrical & Computer Engineering

Entity-Relationship Diagram (ERD) & Database Schema Design

SubTrack — Subscription and Free Trial Manager

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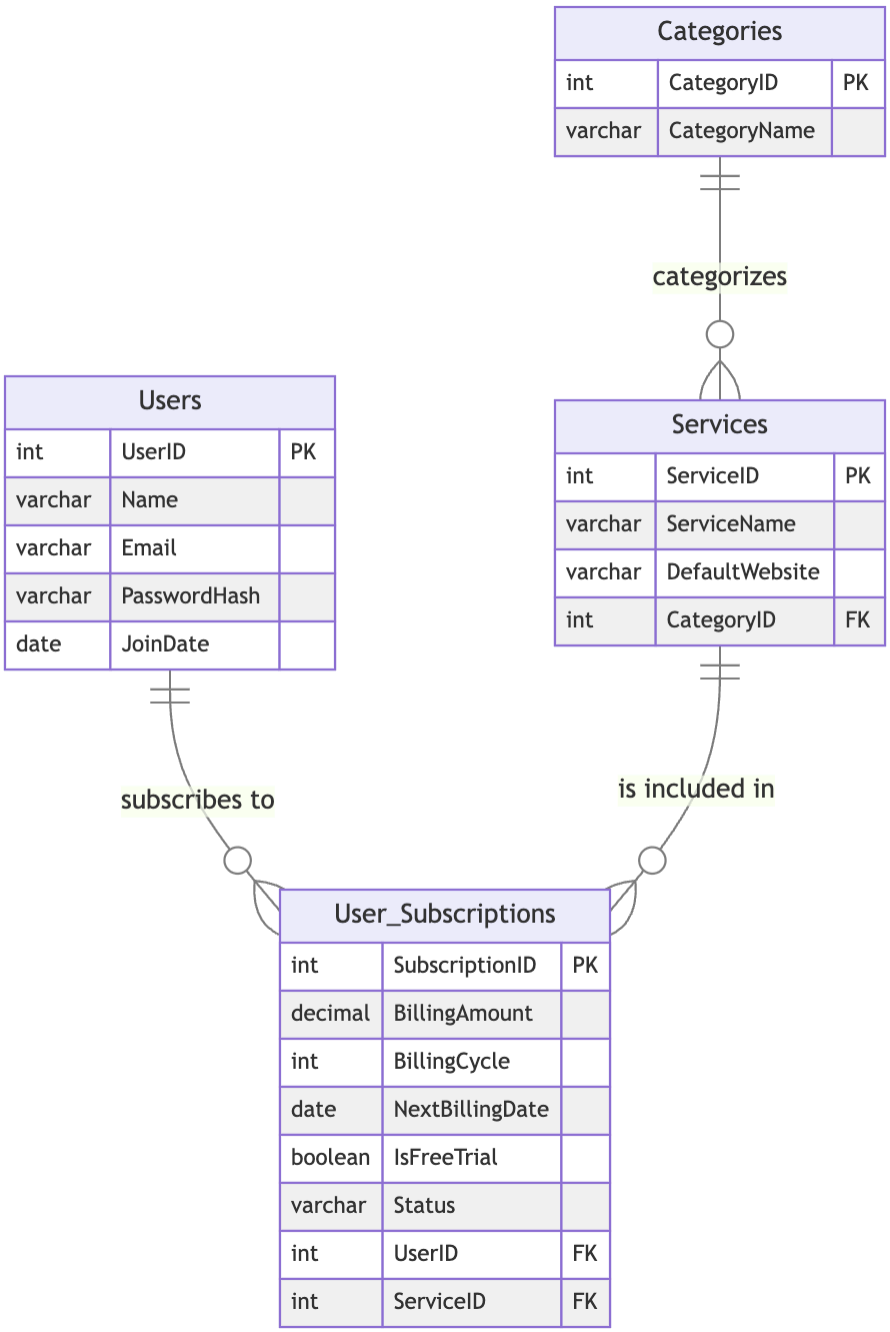
1. Entity-Relationship Diagram (ERD)

The diagram below illustrates the complete ERD for SubTrack using Crow's Foot notation. It shows all four entities, their attributes, and the relationships between them with cardinality



Legend: **PK** = Primary Key, **FK** = Foreign Key, **||** = One and only one (Mandatory), **O<** = Zero or many (Optional)

Figure: ERD for SubTrack — Crow's Foot Notation

2. List of Entities & Attributes

The SubTrack database is built on four core entities. Each entity is defined below with all its attributes, data types, and key constraints. Primary keys are underlined and bold; foreign keys are italicized.

Users

Stores registered user account information for authentication and ownership of subscriptions.

Attribute	Key	Data Type	Description
UserID	**PK**	INT (AUTO)	Unique identifier for each user
Name		VARCHAR(100)	Full name of the user
Email		VARCHAR(150)	Unique email used for login
PasswordHash		VARCHAR(255)	Bcrypt-hashed password string
JoinDate		DATE	Date the user registered

Categories

A lookup table that classifies services into logical groups (e.g., Entertainment, Productivity, Cloud Storage).

Attribute	Key	Data Type	Description
CategoryID	**PK**	INT (AUTO)	Unique identifier for each category
CategoryName		VARCHAR(80)	Human-readable category label

Services

A centralized catalog of digital subscription services. Each service belongs to exactly one category.

Attribute	Key	Data Type	Description
ServiceID	**PK**	INT (AUTO)	Unique identifier for each service
ServiceName		VARCHAR(120)	Name of the service (e.g., Netflix)
DefaultWebsite		VARCHAR(200)	Official URL of the service
CategoryID	**FK**	INT	FK linking to Categories.CategoryID

User_Subscriptions

The central junction/fact table. Records each individual subscription (or free trial) a user holds for a particular service.

Attribute	Key	Data Type	Description
SubscriptionID	**PK**	INT (AUTO)	Unique identifier for each subscription record
BillingAmount		DECIMAL(10,2)	Cost per billing cycle in local currency

Attribute	Key	Data Type	Description
`BillingCycle`		`VARCHAR(20)`	Frequency: Monthly, Yearly, Weekly
`NextBillingDate`		`DATE`	Date of the next charge or trial expiry
`IsFreeTrial`		`BOOLEAN`	TRUE if this is a free trial; FALSE otherwise
`Status`		`VARCHAR(20)`	Current state: Active, Canceled, Expired
`UserID`	**FK**	`INT`	FK linking to Users.UserID
`ServiceID`	**FK**	`INT`	FK linking to Services.ServiceID

3. Relationships & Cardinality

The schema contains three explicit One-to-Many (1:N) relationships. All relationships are enforced through foreign key constraints.

From Entity	Cardinality	To Entity	Description
Categories	**1 : N**	**Services**	One category can contain many services, but each service belongs to exactly one category.
Users	**1 : N**	**User_Subscriptions**	One user can hold many subscription records, but each record belongs to exactly one user.
Services	**1 : N**	**User_Subscriptions**	One service can appear in many subscription records, but each record references exactly one service.

Relationship Narrative

Categories → Services (1:N)

A single category (e.g., "Entertainment") can contain multiple services such as Netflix, Spotify, and Disney+. However, each individual service is classified under only one category. This ensures a clean, non-overlapping taxonomy.

Users → User_Subscriptions (1:N)

A single registered user may have multiple active, canceled, or trial subscriptions on record. Each subscription record is privately owned by and linked to exactly one user account, ensuring data privacy and isolation.

Services → User_Subscriptions (1:N)

A single service (e.g., "Adobe Creative Cloud") can be subscribed to by many different users simultaneously. Each subscription record, however, references only one specific service, preventing ambiguity in billing calculations.

4. Foreign Keys

The three foreign key constraints below enforce referential integrity across the schema. They ensure that no subscription can reference a non-existent user or service, and no service can reference a non-existent category.

Child Table	Foreign Key Column	Parent Table	References Column
Services	CategoryID	Categories	CategoryID
User_Subscriptions	UserID	Users	UserID
User_Subscriptions	ServiceID	Services	ServiceID

Constraint Details

- **CategoryID** is a Foreign Key referencing CategoryID in the Categories table. This constraint prevents a service from being assigned to an undefined category.
- **UserID** is a Foreign Key referencing UserID in the Users table. This constraint ensures every subscription is tied to a valid, registered user account.
- **ServiceID** is a Foreign Key referencing ServiceID in the Services table. This constraint ensures every subscription entry points to a valid service in the catalog.

5. Logical Schema

The logical schema is expressed in standard relational notation. Primary Keys [PK] are underlined; Foreign Keys [FK] show the referenced table. All four tables are independent at the entity level and linked exclusively through foreign key references.

Table Name	Schema Definition
Users	Users(**UserID** [PK], Name, Email, PasswordHash, JoinDate)
Categories	Categories(**CategoryID** [PK], CategoryName)
Services	Services(**ServiceID** [PK], ServiceName, DefaultWebsite, *CategoryID* [FK → Categories])
User_Subscriptions	User_Subscriptions(**SubscriptionID** [PK], BillingAmount, BillingCycle, NextBillingDate, IsFreeTrial, Status, *UserID* [FK → Users], *ServiceID* [FK → Services])